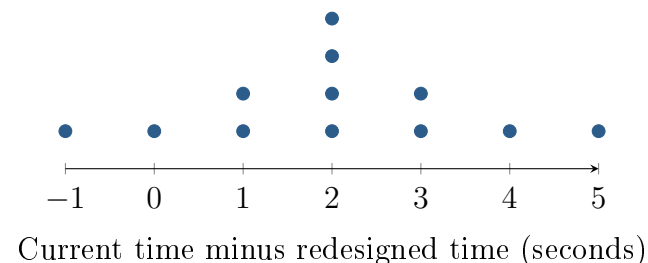


Two Interfaces, Twelve Specialists

Unit 4 · Topic 4.2 · THE PROBLEM

Suppose a hospital randomly samples 12 of its 180 billing specialists. Each does the same task on the current computer screen and, one week later, on a redesigned screen. Everyone uses the current screen first; there is no comparison group. Let d = current – redesigned time in seconds. The plot shows the differences; $\bar{d} = 2.000$, $s_d = 1.6514$, and the 95% critical value is $t^* = 2.201$.



Priya: “Use one sample of paired differences. Doing the tasks in a fixed order limits a cause-and-effect claim.”

Mateo: “Two columns require two independent samples. A positive interval proves the redesign caused faster work.”

FIRST TAKE — one sentence before you work the parts

Which seems more useful here: each person’s change in time or two separate groups of times? Explain in one sentence.

- DOK 1** (a) How many paired differences are there, and what does a positive difference mean?
- DOK 2** (b) Use the supplied summaries to calculate the 95% interval $\bar{d} \pm t^*(s_d/\sqrt{n})$. Show one calculation.
- DOK 3** (c) In 3–4 sentences, choose an analysis and justify it from the linked measurements. Interpret the interval for the 180 specialists, then explain why practice or the fixed order limits a causal claim.

Self-paced bonus problem. Work (a)–(c) from this sheet; turn it in whenever you finish.